

Overview

Interactive Ghana field-course journey. The player moves through 16 course stops in order, reads a short field note at each stop, collects a token, and then uses all tokens to build a final systems map showing that WASH is shaped by more than technical infrastructure.

Dimensions of WASH

Core WASH: Water, Sanitation, Hygiene

Health & Wellbeing

How WASH affects disease, stress, dignity, mental health, physical strain, nutrition, and access to care.

Infrastructure & Technology

Pipes, pumps, toilets, latrines, treatment plants, testing tools, transport systems, maintenance, and design choices.

Governance, Finance & Institutions

Policies, agencies, tariffs, funding, accountability, regulation, public/private roles, and who is responsible for keeping systems working.

Culture, History & Place

Sacred sites, local beliefs, colonial history, national identity, land, memory, and how people give meaning to water, sanitation, and development.

Gender, Equity & Safety

Unequal WASH burdens, women's labour, safety, GBV, menstruation, class, age, access, dignity, and who is included in decision-making.

Climate, Environment & Ecosystems

Flooding, drought, forests, watersheds, pollution, biodiversity, land use, climate variability, and environmental protection.

Livelihoods & Economy

Farming, markets, household income, costs of water/sanitation, productivity, tourism, food systems, and economic survival.

Data, Knowledge & Collaboration

Water testing, monitoring, research, community knowledge, storytelling, photo-elicitation, peer learning, partnerships, and knowledge mobilization.

Mapping WASH: A Systems Journey Through Ghana

Welcome to Mapping WASH, an interactive field-journal game based on SYDE 599: Systems Studies in Water, Environment, and Health. This course explores water, sanitation, and hygiene (WASH) through a mix of lectures, readings, field visits, guest speakers, community conversations, and personal reflection.

In this journey, WASH is not treated as only pipes, pumps, toilets, or treatment plants. Those physical pieces matter, but they are only the hardware of the system. Just as important is the software: governance, financing, education, culture, trust, maintenance, behaviour change, community ownership, and the people responsible for keeping systems working. Each location on the route shows how WASH connects technical systems with health, politics, history, gender, climate, livelihoods, and lived experience.

You will move through the course route stop by stop. At each location, you will open a short field-journal card based on a real course activity, site visit, lecture, reading, or reflection. Each card includes a key image, a collectible token, and several systems lenses that show how that stop connects to the larger WASH system.

How to Play:

Walk along the route and visit each unlocked stop in order.

At each stop:

Read the field journal card to read what happened there.

Look for the systems lenses to see what parts of WASH the stop connects to.

Collect the token before moving to the next location.

At the end, use the tokens to build a WASH systems map.

This is not a complete map of Ghana's WASH system. It is a learning journey through the places, people, and ideas that changed how I understand WASH: from a technical service to a complex system shaped by infrastructure, institutions, culture, health, environment, and everyday life.

Locations Visited

1. Ashesi Orientation + Intro to Ghana

The course began before we entered the field. Through the pre-departure WASH materials and our orientation at Ashesi University, we were introduced to the cultural expectations, language, and relationships that would shape how we learned throughout the course. Orientation included

learning about Ashesi, Ghanaian culture, basic Twi words, and how to approach field visits with respect, curiosity, and awareness of context.

This first stop helped me understand that WASH in Ghana cannot be reduced to pipes, pumps, toilets, or treatment plants. The introduction to WASH in Ghana showed that water, sanitation, and hygiene are managed through a complex mix of government agencies, NGOs, private companies, communities, and traditional authorities. It is not one system controlled by one actor; it is a network of responsibilities, gaps, incentives, histories, and relationships [1]. That was important because it framed every later site visit as part of a larger system.

The biggest idea I took from this stop was the difference between WASH “hardware” and “software.” Hardware is the physical infrastructure people usually notice first: boreholes, pipes, pumps, toilets, tanks, drains, treatment plants, roads, and testing tools. Software is less visible but just as important: governance, financing, maintenance, trust, behaviour change, education, monitoring, community involvement, and cultural acceptance. The WASH reading made it clear that water, sanitation, and hygiene depend on each other, and that infrastructure alone cannot deliver health benefits if people cannot use, maintain, trust, or access it properly [2].

This changed how I wanted to move through the course. Instead of asking only, “What infrastructure exists here?”, I started asking, “Who maintains it? Who pays for it? Who trusts it? Who is excluded from it? What cultural or political factors shape whether it works?”

[1] J. Amengor, “Introduction to WASH in Ghana,” SYDE 599 pre-departure video/lecture, 2026.

[2] London School of Hygiene & Tropical Medicine, “What are water, sanitation and hygiene (WASH) interventions and why are they important?” Water, Sanitation and Hygiene Study Guide, University of London, 2017.

Location icon: book

Key image:



Photo at the Ashesi sign on our first day

Token: Field Passport

WASH focus: Cross-cutting

Systems lenses: Data, Knowledge & Collaboration; Culture, History & Place; Governance, Finance & Institutions

2. Introduction to Water Security + Faircap Water Quality Activity

Our first classroom session began with water at the source. We learned about source water protection, water quality, contamination pathways, and multiple-barrier treatment. During the Faircap activity, the idea of “safe water” became more concrete. Clear water can still carry risks, so testing matters because it helps reveal what people cannot see.

The F-diagram helped me picture how contamination moves. Fecal matter can travel through fingers, flies, fields, fluids, and food before it reaches another person. Each WASH action becomes a barrier in that chain. Toilets separate waste from people. Handwashing interrupts contamination on hands. Safe storage protects water after collection. Treatment reduces pathogens. Cairncross et al. explain that water quality, sanitation, and hygiene are all connected to diarrheal disease prevention, which shows why these barriers have to work together [1].

Multiple-barrier treatment showed that safe water has to be protected at every step from source to consumer. This can include source protection, treatment, storage, and monitoring. The WHO describes drinking-water safety through a risk-management approach that follows water from catchment to consumer, instead of focusing only on the final point of use [2].

This stop changed how I thought about water quality. I started to see safe water as something that has to be protected, tested, stored, and trusted. A source can be labelled improved and still fail people if it is too far away, unreliable, contaminated, or difficult to use every day. Rosinger and Young’s framing of household water insecurity helped me connect this to availability, accessibility, use, reliability, acceptability, and health [3].

[1] S. Cairncross, C. Hunt, S. Boisson, K. Bostoen, V. Curtis, I. C. H. Fung, and W. P. Schmidt, “Water, sanitation and hygiene for the prevention of diarrhoea,” *International Journal of Epidemiology*, vol. 39, no. suppl_1, pp. i193–i205, 2010.

[2] World Health Organization, *Guidelines for Drinking-water Quality: Fourth Edition Incorporating the First and Second Addenda*. Geneva, Switzerland: World Health Organization, 2022.

[3] A. Y. Rosinger and S. L. Young, “The toll of household water insecurity on health and human biology: Current understandings and future directions,” *WIREs Water*, vol. 7, no. 6, 2020.

Location icon: test strip

Key image:



Image of petri plates after water quality test

Token: Test Strip

WASH focus: Water

Systems lenses: Health & Wellbeing; Infrastructure & Technology; Data, Knowledge & Collaboration

3. Lavender Hill / Fecal Sludge Management

At Lavender Hill, we met Frederik Sorenson and Abou, who walked us through a fecal sludge management facility. They explained how waste from trucks can move through filtering, coagulant addition, slow mixing, settling, liquid treatment, and drying of solids. This was the first time sanitation became a full chain in my mind.

Before this visit, I mostly thought about sanitation at the level of the toilet. At this stop, I started to picture everything that happens after the toilet is used. Fecal sludge management includes containment, emptying, transport, treatment, and safe reuse or disposal. WHO uses sanitation safety planning to manage health risks across this full chain, from containment to final end use or disposal [1].

The facility also introduced resource recovery. Waste can still hold value after treatment. Treated water, dried solids, fertilizer, and fuel products can become part of the sanitation economy. Diener et al. explain that resource recovery can create financial incentives for fecal sludge management and shift attention toward useful products from treatment processes [2].

The part that stayed with me most was the political side of the system. The facility had technical value and a clear purpose, yet we learned that its operation was affected by changes in government plans. That made sanitation feel more fragile. A treatment process can be well

designed, but it still depends on institutions, contracts, funding, regulation, and long-term political support. Strande et al. describe fecal sludge management as a systems problem that requires technology, management, and planning together [3].

This stop changed how I understood sanitation infrastructure. I began to see a treatment plant as a meeting point between engineering and governance. Trucks have to arrive. Operators have to be paid. Treatment steps have to work. End products need a safe market. Policies have to survive political change. When one part is missing, the sludge does not disappear. It moves somewhere else in the environment.

[1] World Health Organization, *Sanitation Safety Planning: Manual for Step-by-Step Risk Management for Safely Managed Sanitation Systems*, 2nd ed. Geneva, Switzerland: World Health Organization, 2022.

[2] S. Diener, S. Semiyaga, C. B. Niwagaba, A. Muspratt, J. B. Gning, M. Mbéguéré, D. Ennin, C. Zurbrugg, and L. Strande, "A value proposition: Resource recovery from faecal sludge—Can it be the driver for improved sanitation?" *Resources, Conservation and Recycling*, vol. 88, pp. 32–38, 2014.

[3] L. Strande, M. Ronteltap, and D. Brdjanovic, Eds., *Faecal Sludge Management: Systems Approach for Implementation and Operation*. London, U.K.: IWA Publishing, 2014.

Location icon: sludge truck

Key image:



Token: Sludge Truck

WASH focus: Sanitation

Systems lenses: Infrastructure & Technology; Governance, Finance & Institutions; Health & Wellbeing; Climate, Environment & Ecosystems

4. Arts Market + Kwame Nkrumah Memorial

After visiting a sanitation facility, we moved into a very different kind of learning space: the Kwame Nkrumah Memorial and an arts market. One quote from Nkrumah stayed with me. He argued that progress should be measured through the health of the people, the number of children in school, the quality of education, the availability of water and electricity, and people's

ability to manage their own affairs [1]. This helped me place WASH inside a larger public vision. Water and sanitation are connected to health, education, energy, independence, and the expectations people have for their country.

The arts market brought this idea into everyday life. It was a space of creativity, tourism, bargaining, and income. Walking through the market reminded me that development is also about how people work, trade, create, and support themselves. WASH sits inside that same landscape because water, sanitation, hygiene, health, and livelihoods all shape daily life.

This stop widened the frame of the course. I started to see WASH infrastructure as part of history and public imagination. Systems thinking helped me make that connection. Polaine et al. explain that water security involves people, infrastructure, ecosystems, institutions, and economies, which means WASH problems have to be understood across many parts of society [2].

[1] K. Nkrumah, "Broadcast to the Nation," Dec. 24, 1957, quoted in "Nkrumah Speaks: Nine Memorable Quotes on Ghana's Founders' Day," Face2Face Africa, 2016.

[2] X. K. Polaine, R. Dawson, C. L. Walsh, J. Amezaga, M. Peña-Varón, C. Lee, and S. Rao, "Systems thinking for water security," *Civil Engineering and Environmental Systems*, vol. 39, no. 3, pp. 205–223, 2022.

Location icon: monument

Key image:



Token: Black Star Coin

WASH focus: Cross-cutting

Systems lenses: Culture, History & Place; Livelihoods & Economy; Governance, Finance & Institutions

5. Water for Health Lecture + Local Health Clinic + Paper 1

Before we visited a local health clinic, we learned about the health dimensions of WASH through a lecture and my presentation with Humna on Rosinger and Young's article, "The toll of household water insecurity on health and human biology." Together, these helped me understand that WASH affects health through more than immediate disease. It can shape stress, physical strain, nutrition, chronic illness, and access to care over time.

In the lecture, we discussed health geography, which looks at how health is shaped by place, environment, infrastructure, and social conditions [1]. We also learned that water affects disease in different ways. Some diseases spread through contaminated water, some through poor hygiene when water is limited, some through organisms living in water, and some through vectors like mosquitoes that breed near water [2].

Rosinger and Young expanded this idea by showing that household water insecurity can affect health through contamination, shortage, and excess. They also connect water insecurity to dehydration, diarrhea, malnutrition, psychosocial stress, hypertension, chronic kidney disease, and possible life-course effects [3]. This helped me enter the clinic visit with a broader idea of what "WASH and health" could mean.

At the clinic, these ideas became easier to understand. Malaria was the most common condition they saw, with about 150 to 200 cases each month. The clinic used paper records. Emergency transport depended on taxis, motorbikes, or family drivers. There were also challenges with lighting, road access, delayed reimbursements, and staff shortages.

What stood out to me was how many systems had to work before care could happen. A patient needs transport. The clinic needs staff. The building needs electricity. Records need to be stored and found. Reimbursements need to arrive. Roads need to be usable. Health care depended on much more than medicine.

This stop changed how I understood the health side of WASH. A water or sanitation issue can start at a source, toilet, drain, or mosquito breeding site, then follow people into their homes, routines, bodies, and clinics.

[1] S. Curtis and I. R. Jones, "Is there a place for geography in the analysis of health inequality?" *Sociology of Health & Illness*, vol. 20, no. 5, pp. 645–672, 1998.

[2] S. Cairncross and R. Feachem, *Environmental Health Engineering in the Tropics: An Introductory Text*, 2nd ed. Chichester, U.K.: John Wiley & Sons, 1993.

[3] A. Y. Rosinger and S. L. Young, "The toll of household water insecurity on health and human biology: Current understandings and future directions," *WIREs Water*, vol. 7, no. 6, 2020.

Location icon: first aid symbol

Key image:



Token: first aid cross

WASH focus: Water; Hygiene; Cross-cutting

Systems lenses: Health & Wellbeing; Infrastructure & Technology; Governance, Finance & Institutions; Gender, Equity & Safety

6. Berekuso Spiritual Stream

After the clinic visit, we met Nana Oteng Korankye II, the chief of Berekuso, to receive permission to visit the sacred stream. We were able to move into a space where water was connected to authority, land, spirituality, and respect.

The stream carried more than just water. It carried rules about who could enter, who could give permission, and who could alter the land. The visit made me think about water as something held within relationships. Landowners, traditional councils, government rules, and community expectations all shaped what could happen around the stream.

The hydrosocial cycle helped me make sense of this. It explains that water is shaped by social, political, and economic processes, while also shaping society in return [1]. At Berekuso, the stream shaped local meaning, and local authority shaped how the stream was protected.

Research on sacred groves in Ghana also helped me understand this stop. Esia-Donkoh explains that sacred landscapes can preserve community histories, ancestral knowledge, biodiversity, and local environmental practices [2]. The paper also shows that traditional structures and local actors are important for preserving these places.

This stop made me more cautious about technical solutions. A WASH project in a place like this would need to begin with relationships. It would need to understand who has authority, whose permission matters, what rules already exist, and what the water means to the community. Without that, an intervention could damage the same social system that helps protect the water.

[1] J. Linton and J. Budds, "The hydrosocial cycle: Defining and mobilizing a relational-dialectical approach to water," *Geoforum*, vol. 57, pp. 170–180, 2014.

[2] K. Esia-Donkoh, "Beyond science: Traditional concept of preservation and biodiversity in Ghana: Focus on two traditional areas in Central Region," course reading.

Location icon: stream stone

Key image:



Token: Stream Stone

WASH focus: Water

Systems lenses: Culture, History & Place; Governance, Finance & Institutions; Climate, Environment & Ecosystems

7. WASHKing / Decentralized Sanitation

At WASHKing, we learned that sanitation can be entrepreneurial as well as technical. WASHKing is a decentralized sanitation enterprise that builds toilets and biodigester systems for communities. I noticed the branding right away. "Own Your Throne," the bright colours, and the perfectly delivered pitch made sanitation feel visible, confident, and important to talk about.

Dieudonne's story made the visit feel more personal. He described first encountering a ceramic toilet at age 18 and later building WASHKing without formal business training. That stayed with me because it showed how lived experience can become motivation for design. Sanitation was being treated as a service people could want, invest in, and feel proud of.

Outside, we watched workers lay concrete for the biodigester. We learned that the system uses soak-away technology for liquids and worms for solids. This raised questions for me about groundwater risk, because site conditions matter for any sanitation system that releases liquid

effluent into the surrounding soil [1]. This also made me think about something Frederik said at Lavender Hill: to make a difference, you need to use what you know from training while adapting to the reality of the area. That became one of my main takeaways. A solution can be technically sound and still fail if it does not fit the context where people are expected to use it.

The community visit made the design question feel more human. People, especially women, spoke about privacy and safety. We learned that designing for dignity is a requirement. Nunbogu and Elliott also show that inadequate WASH can expose women and girls to psychological, physical, and sexual violence, which makes safety part of sanitation access rather than a separate issue [2].

This stop changed how I understood sanitation design. A toilet has to work technically, make sense financially, and feel safe to use. People need to want it, trust it, and be able to maintain it.

[1] E. Tilley, L. Ulrich, C. Lüthi, P. Reymond, and C. Zurbrügg, *Compendium of Sanitation Systems and Technologies*, 2nd ed. Dübendorf, Switzerland: Swiss Federal Institute of Aquatic Science and Technology, 2014.

[2] A. M. Nunbogu and S. J. Elliott, "Characterizing gender-based violence in the context of water, sanitation, and hygiene: A scoping review of evidence in low- and middle-income countries," *Water Security*, vol. 15, 2022.

Location icon: toilet

Key image:



Token: toilet

WASH focus: Sanitation

Systems lenses: Infrastructure & Technology; Governance, Finance & Institutions; Livelihoods & Economy; Gender, Equity & Safety

8. Aquaya Guest Lecture + Ashesi Research Symposium

At this stop, Stanley Sam from Aquaya spoke about rural piped water supply systems in Ghana. As I listened, I realized that data is also part of WASH infrastructure. Pipes, pumps, and tanks may be the most visible parts of a water system, but records and financial tracking help determine whether that system can keep working.

Stanley described tools like mWater dashboards, monthly reporting, and key performance indicators. These tools help managers see how systems are performing over time. Without that information, maintenance can become reactive. A system may be losing money, breaking down, or delivering poor service before anyone has a clear picture of the problem.

The lecture also showed why rural water systems can be difficult to manage. Some systems face staff turnover, weak financial records, and missing equipment like meters. These problems affect more than administration. They shape revenue collection, maintenance planning, and long-term reliability. Polaine et al. argue that water data is a foundation for understanding water security systems because it helps connect physical conditions, stakeholder relationships, and decision-making [1].

Later, we attended the Ashesi Research Symposium. Students presented projects on plantain fibres for menstrual pads, machine learning for African languages, health prediction tools, and other topics. Seeing those projects reminded me that innovation was happening all around us. It also made me think about education as a bridge between research and action. A plan for change becomes more powerful when communities are equipped with knowledge they can understand, question, and use to make decisions about their own health, water, and environment.

This stop changed how I understood knowledge in WASH. Better decisions depend on evidence, accountability, and collaboration.

[1] X. K. Polaine, R. Dawson, C. L. Walsh, J. Amezaga, M. Peña-Varón, C. Lee, and S. Rao, "Systems thinking for water security," *Civil Engineering and Environmental Systems*, vol. 39, no. 3, pp. 205–223, 2022.

Location icon: histogram

Key image:

We shall measure our progress by the improvement in the health of our people; by the number of children in school, and by the quality of their education; by the availability of water and electricity in our towns and villages.

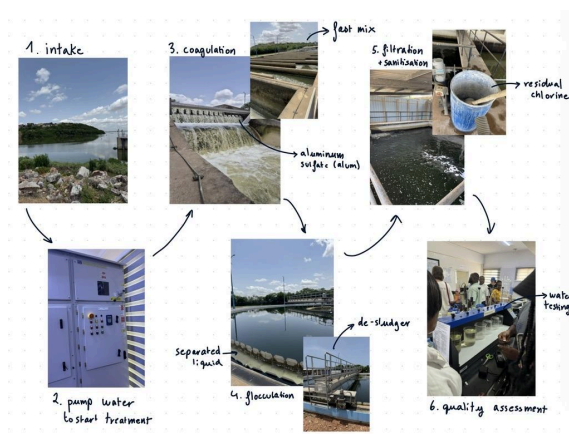
Nkrumah's speech on Ghana's development goals, July 10, 1962.

Token: histogram icon

WASH focus: Water; Cross-cutting

Systems lenses: Data, Knowledge & Collaboration; Governance, Finance & Institutions; Infrastructure & Technology

9. GWCL Weija Treatment Plant + Ghana Water Institute + Paper 2



A process map of the GWCL Weija Treatment Plant showing six stages of centralized water treatment: intake, pumping, coagulation, flocculation, filtration and sanitisation, and final water quality assessment. The image shows how raw water from the Densu River moves through a sequence of treatment barriers before distribution.

At the GWCL Weija Treatment Plant, we saw centralized water treatment using the Densu River, dammed at Weija, as the source water. This stop allowed us to follow a concrete example of the treatment process from lecture.

The process map shows the treatment sequence from left to right. First, raw water enters through the intake and is pumped into the treatment system. This is where the plant begins turning source water into drinking water, but the source still matters because pollution upstream can increase the treatment burden.

Next, the water moves into coagulation and flocculation. In the graphic, alum is added during coagulation, then the water enters fast mixing. This helps small particles clump together. During flocculation, the mixing slows down so larger flocs can form and settle. The de-sludger in the map shows that treatment also produces waste that has to be removed and managed.

Conventional treatment commonly uses coagulation, flocculation, clarification, filtration, and disinfection to reduce risk in drinking water [1].

The final part of the map shows filtration and sanitisation, followed by quality assessment. Filtration removes remaining particles. Chlorine is then added so there is residual protection as water moves through the distribution system. Testing at the end checks whether the treated water meets safety expectations before it reaches users. WHO describes safe drinking water through a risk-management approach that follows the whole system from catchment to consumer [2].

The plant also showed how dependent centralized treatment is on other systems. Power outages can affect pumps and were described as a common cause of misalignment, which was one of the main operational challenges at the plant. Equipment needs maintenance, and workers need the training to notice problems before they become system failures.

The Ghana Water Institute made that point clearer. The institute trains up to 150 water workers per year and provides housing during training, which showed me that education is part of water infrastructure. A treatment plant needs chemicals, pumps, and filters, but it also needs skilled people who understand how to operate, monitor, and repair the system. Training is what helps turn a technical process into a service people can rely on.

A student presentation on Adams and Nyantakyi-Frimpong helped me connect this technical visit to broader urban vulnerability. Their photovoice study focuses on Old Fadama, an informal settlement in Accra where residents experience recurrent flooding. Participants used photographs and interviews to document how flooding affects physical health, mental health, homes, and livelihoods. The authors show that flood risk is shaped by drainage, housing, and political neglect, with impacts that differ by gender, age, and income [3]. This reminded me that treated water is only one part of urban water security. People also need safe housing, working drainage, and institutions that respond before flooding becomes a health crisis.

[1] Environmental Protection Agency, *Water Treatment Manual: Coagulation, Flocculation and Clarification*. Wexford, Ireland: EPA, 2002.

[2] World Health Organization, *Guidelines for Drinking-water Quality*, 4th ed. Geneva, Switzerland: WHO, 2017; World Health Organization, "Water safety plans," in *Guidelines for Drinking-water Quality*.

[3] E. A. Adams and H. Nyantakyi-Frimpong, "Stressed, anxious, and sick from the floods: A photovoice study of climate extremes, differentiated vulnerabilities, and health in Old Fadama, Accra, Ghana," *Health & Place*, vol. 67, Art. no. 102500, 2021.

Location icon: pipe valve

Key image:



Token: Pipe Valve

WASH focus: Water

Systems lenses: Infrastructure & Technology; Health & Wellbeing; Governance, Finance & Institutions; Data, Knowledge & Collaboration

10. Accra School of Hygiene + Community Visit

At the Accra School of Hygiene, we learned about the long history of sanitation education in Ghana. The school opened in 1925 to train sanitation inspectors and has since expanded to over 500 students studying environmental safety, sanitation, and occupational health.

We also heard about student research on community education campaigns, banana peel as a coagulant, and upcycled materials. These projects connected hygiene and sanitation to local problem-solving. They also showed how technical ideas become more useful when they are paired with education.

In Korle-Gonno, we received permission from the chief before entering the community. We visited schools and a library, saw a borehole water installation, and met a hygiene club. The students were learning about handwashing, WASH-related illness, and sachet bag recycling. Hygiene promotion focuses on everyday practices that reduce disease, especially through safe hygiene behaviours like handwashing with soap [1].

A young girl explained to us that handwashing reduces disease. This connects directly to the F-diagram. Fecal contamination can move through hands and food before reaching another person, and handwashing blocks that pathway [2]. At the community library, Belinda explained that children wash their hands before and after touching books, remove their shoes for cleanliness, and use the library as a safe learning space. This was one of the clearest examples of the software side of WASH. Infrastructure matters, but education helps people use and protect it.

1] London School of Hygiene & Tropical Medicine, "What are water, sanitation and hygiene (WASH) interventions and why are they important?" *Water, Sanitation and Hygiene Study Guide*, University of London, 2017.

[2] S. Cairncross, C. Hunt, S. Boisson, K. Bostoen, V. Curtis, I. C. H. Fung, and W. P. Schmidt, "Water, sanitation and hygiene for the prevention of diarrhoea," *International Journal of Epidemiology*, vol. 39, no. suppl_1, pp. i193–i205, 2010.

Location icon: handwashing sign

Key image:



Token: bar of soap

WASH focus: Hygiene; Sanitation; Water

Systems lenses: Health & Wellbeing; Data, Knowledge & Collaboration; Culture, History & Place; Infrastructure & Technology

11. Cape Coast Castle

At Cape Coast, we visited Cape Coast Castle and learned about its role in the trans-Atlantic slave trade. The castle is part of the group of forts and castles along Ghana's coast that UNESCO recognizes as a World Heritage site. These sites were built between the fifteenth and eighteenth centuries as fortified trading posts along the Gold Coast, and they became tied to European trade, colonial power, and the forced movement of enslaved Africans across the Atlantic [1].

Cape Coast Castle changed hands through European competition on the coast. It was first developed by the Swedes in the seventeenth century and later came under British control. Over time, it became one of the major British trading posts on the Gold Coast. Its role also changed as trade shifted more deeply into the capture, confinement, and sale of enslaved Africans [2].

The most striking part of the visit was the physical contrast inside the castle. A church was built directly above the holding dungeons, with an opening between the two spaces. From the male dungeon, enslaved Africans could barely see the church door, while people worshipping above

could hear the suffering below. Our guide described it as a place where “heaven” was built for the colonizers and “hell” was built for the ancestors of the land who were enslaved. That image stayed with me. It showed how religion, power, and violence could exist in the same structure, separated by only a floor.

This stop connected to WASH through history and power. The same landscapes where we were learning about water, sanitation, health, and development were also shaped by colonial rule and inherited institutions. It reminded me that today’s systems have histories. Inequality is built through decisions about land, labour, infrastructure, and whose lives are valued.

[1] UNESCO World Heritage Centre, “Forts and Castles, Volta, Greater Accra, Central and Western Regions,” UNESCO World Heritage List, 1979.

[2] Ghana Museums and Monuments Board, “Cape Coast Castle, Cape Coast (1653),” GMMB.

Location icon: castle

Key image:



Token: Shell

WASH focus: Cross-cutting

Systems lenses: Culture, History & Place; Governance, Finance & Institutions; Gender, Equity & Safety

12. CEWEFIA / Gender and WASH + Community Visit + Paper 3

At CEWEFIA, we saw how gender, WASH, livelihoods, food safety, and coastal ecosystems connect through fish processing. CEWEFIA works with coastal communities through a blue economy lens. Their work supports improved processing, women’s leadership, and alternative livelihoods. The main idea was that when coastal communities are trained and women leaders are supported, ecosystems and livelihoods have a better chance of surviving together.

CEWEFIA's work also connected fish processing to women's rights and protection from gender-based violence. Nunbogu and Elliott explain that WASH-related gender-based violence can be structural, psychological, sexual, and physical [1]. The structural and psychological parts stood out most here. Women's safety and dignity were connected to income, market pressure, and the conditions where food was processed.

In the processing centre, we saw improved ovens, wet rooms, and separate drying areas. We also heard about hygiene expectations, such as handwashing and no jewelry during processing. These details showed how food safety depends on daily routines. The Codex code for fish and fishery products also treats safe fish handling as a full chain, covering production, processing, storage, transport, and sale [2].

The community visit showed why safer processing is difficult to sustain. We saw fish drying on nets where animals could step, fish waste nearby, and women working while also caring for children. When we spoke with fishmongers, what stayed with me was their disappointment that we had seen those conditions, because they explained that it did not reflect the practices they follow. They explained that improved methods can cost more. If they have to charge more for cleaner or more safely processed fish, customers may still choose cheaper fish from people using less safe methods. FAO has also reported that many women fish processors in Ghana still use traditional smoking methods that expose them and their children to smoke and can limit food safety outcomes [3].

This stop changed how I understood food safety. Safer WASH practices have to protect health without punishing the women whose livelihoods already operate under pressure. Skills training, savings systems, and market incentives matter because they help safer practices become realistic.

[1] A. M. Nunbogu and S. J. Elliott, "Characterizing gender-based violence in the context of water, sanitation, and hygiene: A scoping review of evidence in low- and middle-income countries," *Water Security*, vol. 15, Art. no. 100113, 2022.

[2] Codex Alimentarius Commission, *Code of Practice for Fish and Fishery Products*. Rome, Italy: FAO/WHO, 2020.

[3] FAO, "From Smoke to Safety: How FAO Is Transforming Fish Processing in Ghana," United Nations Ghana, 2026.

Location icon: fish

Key image:



Token: Fish

WASH focus: Water; Sanitation; Hygiene

Systems lenses: Gender, Equity & Safety; Health & Wellbeing; Livelihoods & Economy; Culture, History & Place

13. Sacred Forest

At Nsofa Forest, we visited a sacred grove with Professor Kobina Esia-Donkoh from the Department of Population and Health at the University of Cape Coast, alongside members of the Eshirow Community. The community chief gave us his blessing to walk through the grove. This made the visit feel guided by local authority, respect, and community knowledge.

During the visit, we learned about the cultural and spiritual beliefs connected to the forest. We also participated respectfully in a ritual where alcohol was poured at a place connected to the god of the forest. Permission mattered here. The forest was a living place with rules, memory, and responsibility.

In a paper Professor Esia-Donkoh wrote about sacred groves in Ghana, he explains that groves are patches of vegetation preserved through local knowledge for cultural, religious, and ecological purposes [1]. He identifies Nsofa as a grove in Eshirow connected to both a god and ancestral memory. The paper also describes Nsofa as preserved and owned by the Anona Clan [1].

The spiritual meaning of the grove helped create rules around access, permission, and use. Esia-Donkoh explains that sacred groves are often protected through taboos, rituals, and traditional authority. In some groves, people could not farm, hunt, or cut trees. Herbs and dead wood could only be collected after permission from the priest or local authorities [1].

This helped me see spirituality as part of environmental protection. In the paper, ancestors are described as helping communities with rains and good harvests, and Esia-Donkoh notes that

Nsofa held similar significance [1]. Protecting the grove also protected trees, plants, and biodiversity. The belief system gave people a reason to care for the forest across generations.

The visit also showed why knowledge-sharing matters. Education should include community knowledge so different knowledge systems can share space instead of creating conflict. Going forward, this stop makes me want to approach conservation and WASH work more carefully. Local knowledge systems should be included in the solution.

[1] K. Esia-Donkoh, "Beyond science: Traditional concept of preservation and biodiversity in Ghana: Focus on two traditional areas in Central Region," Department of Population and Health, University of Cape Coast, Ghana.

Location icon: tree

Key image:



Token: bamboo

WASH focus: Water; Cross-cutting

Systems lenses: Culture, History & Place; Climate, Environment & Ecosystems; Governance, Finance & Institutions; Data, Knowledge & Collaboration

14. Kakum National Park

At Kakum National Park, we went on a forest hike and crossed the canopy walkway. From above the forest floor, the trees felt like part of the WASH system. Forests help regulate water, protect watersheds, and support biodiversity. They also create livelihoods through conservation and tourism. This is connected to Polaine et al.'s systems thinking approach, which explains that water security depends on relationships among people, infrastructure, ecosystems, economies, and governance [1].

At the next stop, we'll learn more about climate-resilient WASH. Kakum helps introduce that idea because it shows that water systems depend on climate protection. Forests help protect watersheds, support rainfall patterns, reduce erosion, and create healthier ecosystems. This means climate-resilient WASH has to include land use, biodiversity, and environmental protection.

[1] X. K. Polaine, R. Dawson, C. L. Walsh, J. Amezcaga, M. Peña-Varón, C. Lee, and S. Rao, "Systems thinking for water security," *Civil Engineering and Environmental Systems*, vol. 39, no. 3, pp. 205–223, 2022.

Location icon: canopy leaf

Key image:



Token: Canopy Leaf

WASH focus: Water; Cross-cutting

Systems lenses: Climate, Environment & Ecosystems; Culture, History & Place; Livelihoods & Economy; Health & Wellbeing

15. Cape Coast Activity + Technical Recap + WaterAid Guest Lecture

This stop brought together our Cape Coast photo-elicitation activity and the WaterAid guest lecture with Ing. Richard Agbo Ntibrey on WASH and climate resilience. The photo activity asked us to look at three questions: how gender and age affect access to water, where we saw sanitation and hygiene gaps, and how climate change contributes to water insecurity.

One of my photos showed a goat walking across fish drying nets. I first took the photo because the goat caught my attention, but later I noticed the WASH issue inside the image. The goat was walking on the same nets where fish were being dried. That photo connected food safety, hygiene, and livelihoods in one scene.

Michael shared a photo of a backed-up gutter. His reflection connected the stagnant water to mosquitoes and illness. The image showed how poor drainage can become a health risk, especially when flooding or heavy rainfall puts more stress on the system.

Teagan's reflection stayed with me most. She described a photo from the bus route and pointed out that women do not have the same freedom as men to stop and urinate while travelling. Men can often step away more easily, while women have to think about privacy, safety, and social judgment. That reflection directly answered the prompt about gendered access and showed how sanitation inequality can appear in a very ordinary travel moment.

The WaterAid lecture helped connect these observations to climate-resilient WASH. Ing. Ntibrey explained that climate change can damage WASH systems through flooding, drought, and storms. Flooding can contaminate water sources, drought can dry up boreholes, and storms can damage infrastructure [1]. These risks become more dangerous when communities already have weak drainage, limited sanitation, or unreliable water access. He also emphasized that climate-resilient WASH has to be accessible for all users. A system cannot be considered resilient if children, older adults, people with disabilities, or women managing care work cannot safely use it.

The lecture also emphasized that climate-resilient WASH begins with assessment and then response. Communities need to understand their risks before solutions are designed. UNICEF and the Global Water Partnership describe climate-resilient WASH as planning that accounts for climate hazards, strengthens services, and supports adaptation over time [2].

[1] R. A. Ntibrey, "WASH and Climate Change," WaterAid guest lecture, SYDE 599, 2026.

[2] Global Water Partnership and UNICEF, *Strategic Framework for WASH Climate Resilient Development*, 2022 ed. Stockholm, Sweden: GWP and UNICEF, 2022.

Location icon: camera

Key image:



Token: Camera

WASH focus: Cross-cutting

Systems lenses: Data, Knowledge & Collaboration; Governance, Finance & Institutions; Climate, Environment & Ecosystems; Health & Wellbeing

16. Tetteh Quarshie Cocoa Farm + Paper 4

At Tetteh Quarshie Cocoa Farm, we learned about cocoa plants, the history of cocoa cultivation in Ghana, and the way farming connects climate, government, and livelihoods. The farm is widely associated with Tetteh Quarshie, who returned from Fernando Po, now Bioko Island in Equatorial Guinea, with cocoa pods in 1879 and planted them in Akuapim-Mampong [1]. Ghana Tourism Authority describes this farm as the first successful cocoa plantation in the country and an important part of Ghana's cocoa heritage [1].

At this stop, we learned that cocoa is a sensitive agricultural system. Cocoa trees need suitable rainfall, shade, soil, and care. The guide explained that cocoa is vulnerable because changes in climate can affect how well the trees grow and produce pods. Research on cocoa farming in Ghana also shows that changing rainfall and temperature can affect cocoa productivity and farmer livelihoods [2].

Cocoa is also tied to Ghana's economy and governance. A USDA report states that Ghana's cocoa sector employs about 800,000 farm families and generates around US\$2 billion in foreign exchange annually [3]. The guide explained that cocoa farmers sell into a state-structured system. COCOBOD regulates internal cocoa marketing, sets producer prices with government, and works through licensed buying companies [4]. The guide also said that cocoa-growing areas often get better roads because cocoa has to be transported from farms to buyers. This showed me how one crop can shape infrastructure decisions.

Polaine et al.'s systems-thinking argument helped me connect this farm visit back to the whole course. They explain that water security involves people, built environments, natural environments, economies, and the interactions between them [5]. A cocoa farm depends on all of those pieces. It needs rainfall and soil, but it also needs roads, markets, labour, environmental stability, and policy decisions.

[1] Ghana Tourism Authority, "Chocolate Week," *Visit Ghana*, 2025.

[2] E. M. Osei, "Climate Variability and Cocoa Production: The Implications of Micro-Adaptation Measures on Cocoa Farmers' Income in the Ahafo Ano South District," University of Ghana, 2017.

[3] USDA Foreign Agricultural Service, *Ghana - Cocoa Sector Overview - 2025*, Report no. GH2025-0008, 2025.

[4] Ghana Cocoa Board, "Objectives and Functions of the Board," COCOBOD, 2026.

[5] X. K. Polaine, R. Dawson, C. L. Walsh, J. Amezaga, M. Peña-Varón, C. Lee, and S. Rao, "Systems thinking for water security," *Civil Engineering and Environmental Systems*, vol. 39, no. 3, pp. 205–223, 2022.

Location icon: cocoa pod

Key image:



Image of me holding a cocoa pod

Token: Cocoa Pod

WASH focus: Water; Cross-cutting

Systems lenses: Livelihoods & Economy; Climate, Environment & Ecosystems; Culture, History & Place; Governance, Finance & Institutions